

Kubernetes Deployment Assignment

Objective

Deploy a **Flask frontend** and **Express backend** application on a local Kubernetes cluster using **Minikube**.

Project Structure

kubernetes-assignment/

```
|
|
| └─ backend/
|   └─ Dockerfile
|   └─ package.json
|   └─ server.js
|
| └─ frontend/
|   └─ Dockerfile
|   └─ app.py
|
└─ k8s/
    └─ backend-deployment.yaml
    └─ backend-service.yaml
    └─ frontend-deployment.yaml
    └─ frontend-service.yaml
```

Technologies Used

-  Docker

-  Kubernetes
 -  Minikube
 -  Flask (Frontend)
 -  Express.js (Backend)
-

Setup & Deployment Steps

Start Minikube

```
minikube start
```

Configure Docker Environment

```
minikube docker-env | Invoke-Expression
```

Build Docker Images

```
docker build -t backend-app ./backend
```

```
docker build -t frontend-app ./frontend
```

Deploy to Kubernetes

```
kubectl apply -f k8s/backend-deployment.yaml
```

```
kubectl apply -f k8s/backend-service.yaml
```

```
kubectl apply -f k8s/frontend-deployment.yaml
```

```
kubectl apply -f k8s/frontend-service.yaml
```

Verify Deployment

```
kubectl get pods
```

```
kubectl get services
```

Access Application

minikube service frontend-service

Screenshots

☒ Pods Running

(Add screenshot of kubectl get pods)

☒ Services Running

(Add screenshot of kubectl get services)

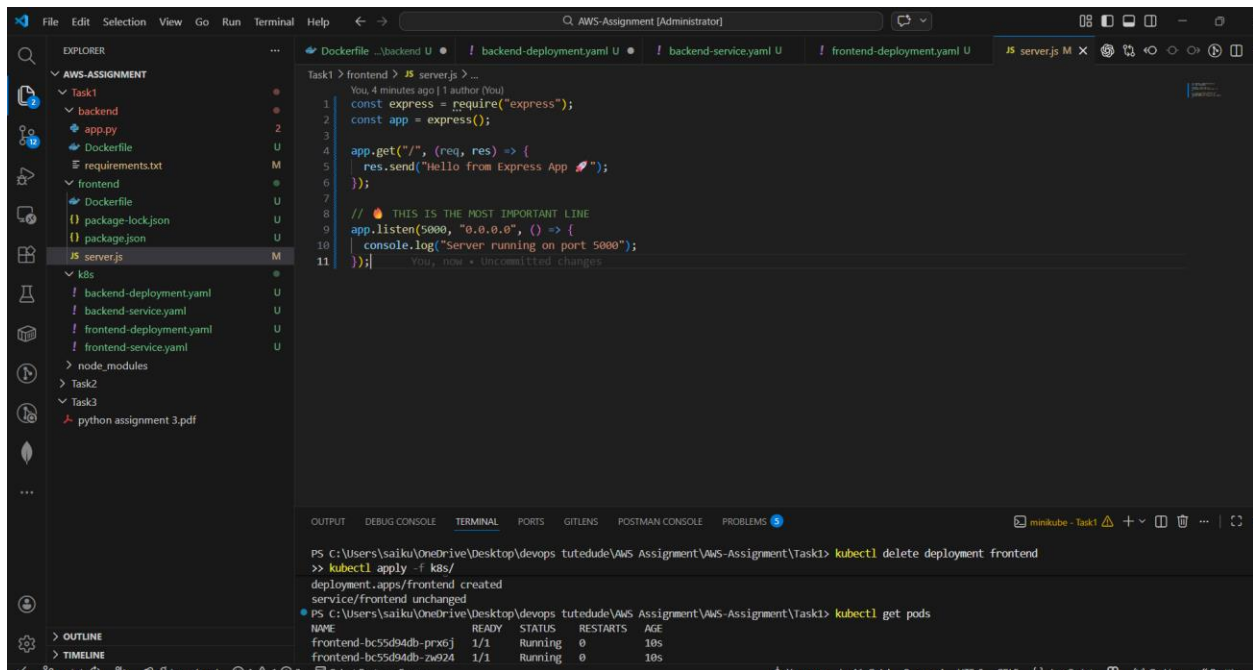
☒ Application Output

(Add browser screenshot)



Hello from Express App 





□ Architecture

User → Frontend (Flask) → Backend (Express API) → Response

- Frontend communicates with backend via service
- Backend handles API requests
- Kubernetes manages scaling and networking

☑ Output

- All pods are running successfully
- Services are exposed using NodePort
- Application is accessible via Minikube